1. Study whether each of the following situations is possible. For the possible cases, give explicit examples of hypothesis classes H_1 and H_2. For impossible cases, if there are any, justify why it cannot occur.

a) $\text{VCdim}(H_1 \cup H_2) < \max(\text{VCdim}(H_1), \text{VCdim}(H_2))$

b) $\text{VCdim}(H_1 \cup H_2) = \max(\text{VCdim}(H_1), \text{VCdim}(H_2))$

c) $\text{VCdim}(H_1 \cup H_2) > \max(\text{VCdim}(H_1), \text{VCdim}(H_2))$

2. Consider H, the class of 3-piece classifiers (signed intervals),

$$H = \{ \, h_{a,b,s} : \mathbb{R} \to \{-1, 1\} \mid a \leq b, \;\; s \in \{-1, 1\} \, \}$$

where

$$h_{a,b,s}(x) = \begin{cases} s, & x \in [a, b] \\ -s, & x \notin [a, b] \end{cases}$$

We know from the seminar class that $\text{VCdim}(H) = 3$.

a) Compute the shattering coefficient $\tau_H(m)$ of the growth function for $m \geq 0$, for hypothesis class H.

b) Compare your result with the general upper bound for the growth function given by Sauer's Lemma and show that the $\tau_H(m)$ obtained at (a) is not equal to the upper bound.

c) Does there exist a hypothesis class H for which the shattering coefficient $\tau_H(m)$ is equal to the general upper bound, over the same or over another domain X? If your answer is yes, provide an example. If your answer is no, provide a justification.

(3) Consider the unknown labeling function $f_u : \mathbb{R} \rightarrow \{0, +1\}$

that labels real points x_i with the binary label $y_i = f_u(x_i) \in \{0, +1\}$.

We consider the concept class C given by the closed intervals

$$[a, a + 10]$$

where $a \in \mathbb{R}$. You are given the following particular labeled training set S:

$$S = \Big\{(-3, +1),\ (-1, 0),\ (0, 0),\ (2, +1),\ (3, +1),\ (5, 0),\ (7, +1),\ (9, +1),\ (11, 0),\ (13, +1)\Big\}.$$

where all points $(x_i, y_i) \in S$ are obtained using f_u, as

$$y_i = f_u(x_i).$$

a) Give a hypothesis $h_{a_S} = \mathbf{1}_{[a_S,\, a_S+10]}$ from C that achieves the minimum error on S.

b) Write an efficient ERM learning algorithm for the concept class C in the general case, for an arbitrary training set S.